

Run a SNOMED CT terminology server on your laptop

Snowstorm 11 and Elasticsearch 8 with Docker: a hands-on workshop for developers

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What you will have by the end



1. Snowstorm running

Elasticsearch and Snowstorm in Docker, reachable on localhost only



2. A release loaded

The International Edition 20261001, imported in about 65 minutes



3. Hands-on with Postman

$lookup, $expand, $validate-code and $translate on our test server



4. The pitfalls, avoided

The ones we hit running Snowstorm for the ARP registries, and today's dry run

Before you start

What runs where, and what your laptop needs

Two containers, and one port you call



- **Both ports bind to 127.0.0.1.** Nothing else on the network can reach them
- **Elasticsearch holds everything.** Delete the volume and you import again
- **Snowstorm is stateless.** Restart it freely, but its search cache starts cold (pitfall 7)

Check your laptop before downloading anything

8 GB

memory for Docker: Docker Desktop: Settings, Resources

20 GB

free disk inside Docker: the loaded index is ~2.6 GB

588 MB

the RF2 zip: International Edition 20261001

- **Docker with Compose v2:** `docker compose version`
- **Linux or WSL2 only:** raise vm.max_map_count (below)
- **Ports 8080 and 9200 free** on localhost
- **curl or Postman** for the FHIR calls
- **The RF2 release from MLDS,** under Malaysia's SNOMED licence

```
# Linux
sudo sysctl -w vm.max_map_count=262144
# Windows + WSL2 (PowerShell)
wsl -d docker-desktop sysctl -w vm.max_map_count=262144
```

Step by step

Five steps from an empty laptop to a read-only FHIR server

Step 1 · Get the release file from MLDS

1 Sign in to MLDS

mlds.ihtsdotools.org, as an affiliate in a member country (Malaysia is one)

2 Download the International Edition

The latest monthly release; keep the file name, it carries the date

3 Keep it zipped, and out of git

Snowstorm reads the zip as is. The content is licensed: never commit or share it

Inside the zip

```
SnomedCT_InternationalRF2_PRODU
  CTION_20261001T120000Z.zip
├─ Snapshot/  ← what we import
│  ├─ Terminology/  concepts, descriptions
│  └─ Refset/       language, maps, history
├─ Full/       every version since 2002
└─ Readme, release_package_information.json
```

Snapshot = the current state of every component, active and inactive. Enough for a terminology server; Full adds the history and takes much longer to import.

Step 2 · Write compose.yaml

```yaml
services:
  elasticsearch:
    image: docker.elastic.co/elasticsearch/elasticsearch:8.19.8
    environment:
      - discovery.type=single-node
      - xpack.security.enabled=false
      - ES_JAVA_OPTS=-Xms${ES_HEAP:-4g} -Xmx${ES_HEAP:-4g}
    volumes: [ es-data:/usr/share/elasticsearch/data ]
    healthcheck: { test: curl -fs localhost:9200/_cluster/health }
    ports: [ "127.0.0.1:9200:9200" ]
  snowstorm:
    image: snomedinternational/snowstorm:11.0.0
    depends_on: { elasticsearch: { condition: service_healthy } }
    environment:
      - JAVA_TOOL_OPTIONS=-Xmx${SNOWSTORM_HEAP:-4g} ...
    command:
      - --elasticsearch.urls=http://elasticsearch:9200
      - --snowstorm.rest-api.readonly=${READONLY:-false}
    ports: [ "127.0.0.1:8080:8080" ]
volumes: { es-data: {} }
```

1. **Published images, pinned**

   Upstream's file builds from source. Pin the version you tested

2. **Localhost-only ports**

   127.0.0.1 keeps both off the workshop Wi-Fi

3. **JAVA_TOOL_OPTIONS**

   The image ignores JAVA_OPTS. Set heaps here, or you get the default

4. **One read-only switch**

   Writable to load, then READONLY=true to serve

Step 3 · Start it, and check both containers

```
# 8 GB+ for Docker
docker compose up -d
# smaller laptop
ES_HEAP=2g SNOWSTORM_HEAP=2g docker compose up -d

curl localhost:8080/version
{ "version" : "11.0.0", ... }

curl -s localhost:9200/_cluster/health | jq .status
"green"

curl -s localhost:8080/codesystems | jq '.items[].shortName'
"SNOMEDCT"      # exists, but empty until step 4
```

36 s

from up to a /version answer, empty volume

Then open

`localhost:8080`
Swagger UI: every native endpoint

`localhost:8080/fhir/metadata`
FHIR CapabilityStatement

Step 4 · Import the RF2 snapshot

1 **Create an import job**

```
curl -i -X POST localhost:8080/imports \
  -H 'Content-Type: application/json' \
  -d '{"branchPath":"MAIN","type":"SNAPSHOT",
       "createCodeSystemVersion":true}'
Location: .../imports/e687cbbf-...
```

2 **Upload the zip to it**

```
curl -X POST -F file=@SnomedCT_InternationalRF2_...20261001T120000Z.zip \
  localhost:8080/imports/<id>/archive
```

3 **Poll until COMPLETED**

```
curl -s localhost:8080/imports/<id> | jq .status
"RUNNING" ... "COMPLETED"
```

65 min

measured for 20261001 on a MacBook with 6 GB for Docker; upstream says 30 to 60

While it runs

`docker compose logs -f snowstorm`

shows batches of 5,000 concepts, descriptions and relationships. Leave it alone: a restart means a clean volume and a fresh start

Step 5 · Switch to read-only, and keep the data

```
READONLY=true docker compose up -d
# recreates Snowstorm only; Elasticsearch and the data stay

curl -s -o /dev/null -w '%{http_code}' -X POST localhost:8080/imports
403
```

What keeps your data

`docker compose stop`	stop; data kept
`docker compose down`	remove containers; data kept
`docker compose down -v`	deletes the index: import again



Why bother on a laptop?

Because your application will meet a read-only server everywhere else. If it only works against a writable one, you find out in production

Call it like an application

Four FHIR operations, with real responses from the dry run

$lookup: what does this code mean?

```
GET /fhir/CodeSystem/$lookup?system=http://snomed.info/sct&code=322236009
```


```
{ "resourceType": "Parameters", "parameter": [
  { "name": "display",
    "valueString": "Acetaminophen 500 mg oral tablet" },
  { "name": "version", "valueString":
    "http://snomed.info/sct/900000000000207008/version/20261001" },
  { "name": "designation", "part": [ ...
      "valueString": "Paracetamol 500 mg oral tablet" ] },
  { "name": "property", "part": [
      { "name": "code", "valueCode": "762949000" },
      { "name": "code-display",
        "valueString": "Has precise active ingredient" },
      { "name": "value", "valueCode": "387517004" } ] },
  ... ] }
```


display

US English by default. Accept-Language: en-GB returns Paracetamol 500 mg oral tablet

version

The edition and release that answered: pin it in tests

valueCode

11.0.0 changed properties from valueString: the one breaking change

$expand: type-ahead search inside a value set

```
GET /fhir/ValueSet/$expand?url=http://snomed.info/sct?fhir_vs=ecl/%3C%3C763158003
    &filter=parac&count=5
```


```
{ "resourceType": "ValueSet",
  "expansion": {
    "total": 386,
    "contains": [
      { "system": "http://snomed.info/sct",
        "code": "90332006",
        "display": "Acetaminophen-containing product" },
      { "code": "35324004",
        "display": "Paraldehyde-containing product" },
      { "code": "424102008", "display":
        "Acetaminophen- and aspirin-containing product" },
      ... ] } }
```


29.5 s first call after a restart

1.1 s same call again

`%3C%3C` is <<. A raw < gets HTTP 400

`count` sets the page size; _count is ignored: all 386 came back

$validate-code and $translate: check and repair codes

$validate-code

```
GET /fhir/ValueSet/$validate-code
  ?url=http://snomed.info/sct?fhir_vs=ecl/%3C%3C763158003
  &system=http://snomed.info/sct&code=322236009
```

```
{ "name": "result", "valueBoolean": true },
{ "name": "display",
  "valueString": "Acetaminophen 500 mg oral tablet" }
```

```
{ "name": "result", "valueBoolean": false },
{ "name": "message", "valueString": "The provided code
  'http://snomed.info/sct#387517004' was not found
  in the value set ...' }
```

A substance in a medicinal-product field fails, with a reason

$translate

```
GET /fhir/ConceptMap/$translate
  ?url=http://snomed.info/sct?fhir_cm=900000000000527005
  &system=http://snomed.info/sct&code=373988004
```

```
{ "name": "result", "valueBoolean": true },
{ "name": "match", "part": [
  { "name": "equivalence", "valueCode": "equal" },
  { "name": "concept", "valueCoding": {
      "system": "http://snomed.info/sct",
      "code": "321241004" } } ] }
```

SAME AS map: the replacement for an inactivated code

Hands-on with Postman

On our shared test server, api-test.mysnomed.com, so nobody waits for an import

The test server: what it serves, and the house rules

`https://api-test.mysnomed.com`

What it serves

SNOMED CT International
20260901 · 384,050 active concepts

LOINC extension
20260321 · code system SNOMEDCT-LOINCEXT

KKM health facilities
20251231.1 · 5,165 codes, 9 ValueSets

Snowstorm 10.12.2
read-only, FHIR at /fhir



One IP for the whole room

Send requests one at a time. No Collection Runner



Credentials from the facilitator

Put them in Current value only. Never export or screenshot them



Responses are cached for 24 h

For language, use displayLanguage=, not the Accept-Language header



Read-only

Nothing you send can change it. Experiment freely

Set up Postman in four steps

1 **Import both files**

File, Import, then drop in the collection and the environment

2 **Select the environment**

Top right: api-test.mysnomed.com

3 **Fill in the credential**

Type the password from the facilitator in Current value only

4 **Send 0.1 Server version**

Expect 200 and "version": "10.12.2". A 401 means step 2 or 3

Environment: api-test.mysnomed.com

Variable	Current value
baseUrl	https://api-test.mysnomed.com
sct	http://snomed.info/sct
kkm	https://mysnomed.com/fhir/draft
username	workshop
password	•••••••• (secret)

Collection: 52 requests in 10 folders

0 · Check you are connected
1 · Look up a code
2 · Search ($expand)
3 · Validate a code
4 · Inactive codes
5 · KKM facilities
6 · Native SNOMED API
7 · Pitfalls, on purpose
8 · Pharmacy examples
9 · Clinical examples

Anatomy of one request in Postman

```
GET  {{baseUrl}}/fhir/ValueSet/$expand?url={{sct}}?fhir_vs=ecl/<<763158003&filter=parac&count=10
```

Params	Authorization	Headers	Tests
url {{sct}}?fhir_vs= ecl/<<763158003 filter parac count 10	Inherit auth from parent → Basic Auth {{username}} {{password}}	Accept: application/fhir+json added by the collection, /fhir only	pm.test('found', () => pm.expect(pm.response.json() .expansion.total).to.be.above(0));
The value set is an ECL rule inside url. Postman encodes the <<	Set once on the collection	The native API answers 500 to this header	A green tick means the expected answer

The response pane: read expansion.total and contains[].code / display. Status, time and size sit above it

Exercises 1 to 4: look up and search

1 `1.1`

What is the display of 322236009, and which release answered?

2 `1.2`

Ask for British English. What changes in the display?

3 `2.1 → 2.2`

How many products match 'parac'? Now try 'pa'. Why the difference?

4 `2.3 · 2.4`

<<90332006 gives one total, the 'has active ingredient' refinement another. Why do they differ?

Exercises 5 to 8: validate, history, local content

5 `3.1 · 3.2`

Validate 322236009, then 387517004, against medicinal products. What does the failure say?

6 `4.1 → 4.2`

Is 373988004 active? If not, what replaced it?

7 `5.2 · 5.3`

Find Hospital Kuala Lumpur's facility code. Which state and sector does it record?

8 `Stretch`

Write ECL for every clinical drug that contains amoxicillin. How many are there?

Pharmacy examples: folder 8 of the collection

Req	What it shows	Result on the test server
8.1	Prescribing search: clinical drugs only, matching 'amlodipine'	**79 clinical drugs**
8.2	Every clinical drug containing amoxicillin, co-amoxiclav included	**30**
8.3	Paracetamol oral tablets: ingredient AND dose form	**94**
8.4	Warfarin clinical drugs, searched by the base substance	**0: the salt trap**
8.5	The same, including salts via 738774007 |Is modification of|	**11**
8.6	Every product containing any penicillin (a drug class)	**145**
8.7	Allergy picker: substances matching 'penicillin'	**23**
8.8	Every 'allergy to a penicillin' condition	**26**
8.9	Everything recorded as caused by amoxicillin	**3: allergy, reaction, overdose**
8.10–12	Dose forms 'tablet', routes 'oral', units of presentation	**32 · 1 · 59**
8.13	A clinical drug's definition as a table (Visualize tab)	**9 attributes, 500 mg per tablet**

Clinical examples: folder 9 of the collection

Req	What it shows	Result on the test server
`9.1`	Diagnosis picker: diseases matching 'type 2 diab'	**134**
`9.2`	Symptoms and findings matching 'chest pain'	**38**
`9.3–4`	Is it a valid diagnosis? Hypertension, then chest pain	**true · false**
`9.5`	Parents and children of type 2 diabetes	**1 parent, its children**
`9.6`	Every kind of diabetes mellitus	**119**
`9.7`	Every disorder of the kidney, by finding site	**1,435**
`9.8–9`	SNOMED CT to ICD-10: type 2 diabetes, hypertension	**E11.9 · I10**
`9.10`	Procedures matching 'appendicectomy'	**11**
`9.11`	Body sites matching 'left kidney'	**13**
`9.12`	Observations matching 'blood pressure'	**83**
`9.13`	Lab orders in the LOINC extension: 'hemoglobin a1c'	**8**

From Postman to your code

JavaScript (Node 18+)

```
const base = 'https://api-test.mysnomed.com';
const auth = 'Basic ' + Buffer.from(
  `${process.env.TS_USER}:${process.env.TS_PASS}`
).toString('base64');

const q = new URLSearchParams({
  url: 'http://snomed.info/sct?fhir_vs=ecl/<<763158003',
  filter: 'parac', count: '10',
});
const res = await fetch(
  `${base}/fhir/ValueSet/$expand?${q}`, { headers: {
    Authorization: auth,
    Accept: 'application/fhir+json',
    'Accept-Language': 'en-GB',  // not *
  } });
const { expansion } = await res.json();
```

Python (requests)

```
import os, requests

r = requests.get(
  'https://api-test.mysnomed.com/fhir/ValueSet/$expand',
  params={'url': 'http://snomed.info/sct?fhir_vs=ecl/<<763158003',
          'filter': 'parac', 'count': 10},
  auth=(os.environ['TS_USER'], os.environ['TS_PASS']),
  headers={'Accept': 'application/fhir+json',
           'Accept-Language': 'en-GB'},
  timeout=10)
total = r.json()['expansion']['total']
```

Postman, URLSearchParams and requests encode the ECL. Only a URL you build by hand, or type into curl, needs << written as %3C%3C

When a request fails on the test server

You see	Most likely	Do this
401 Unauthorized	No environment selected, or the password is only in Initial value	Select api-test.mysnomed.com; fill Current value
500 on /version, /codesystems, /MAIN/…	A FHIR Accept header sent to the native API	Send application/fhir+json to /fhir only
500 HAPI-0389 … Accept-Language '*'	The client sends Accept-Language: * (Node fetch does)	Send en or en-GB
500 HAPI-0389 … / by zero	count=0 on $expand	Use count=1 and read expansion.total
400 Bad Request, an HTML page	A raw < in a hand-built URL or curl command	Encode it: << is %3C%3C
200, but total is 0	Inactive or mistyped SCTID in the ECL, or a SNOMED filter under 3 letters	$lookup the code; type 3 letters or more
200, but the other language	A cached response: the cache ignores Accept-Language	Use displayLanguage=en-GB
503 from nginx	Over 20 requests a second from the room's one IP	Wait a few seconds; no Collection Runner
503 from the load balancer	The server is stopped (every night at 21:00 MYT)	Tell the facilitator

Pitfalls

Eight things that cost us time, and how to spot them early

Pitfalls: the symptom you see, and what it means

You see	What is going on	Do this
First start takes many minutes; 'Creating index' every 30 s	Docker's disk is over 90% full, so Elasticsearch refuses to place data (cluster red)	Free space (docker builder prune); keep 20 GB free
The heap is not what you set	The image starts java directly: JAVA_OPTS is ignored	Use JAVA_TOOL_OPTIONS; check 'Picked up' in the log
First search after a restart takes ~30 s	Snowstorm's ECL cache is in memory and starts empty	Warm your app's value sets after every restart
200, but total is 0	The ECL names an inactive concept	Check every value set's size after each release
400 'Unknown SNOMED Concept Id' (11.x) or an empty 200 (10.x)	A mistyped SCTID in the ECL	Same: a minimum-size test per value set
Results look right, the |term| is wrong	Snowstorm does not check |terms| against codes	Never trust a term you did not look up
LOINC or national codes are missing	Unversioned ECL searches the International Edition only	Name the edition: sct/<module>?fhir_vs=...
Acetaminophen, not Paracetamol	US English is the default display language	Send Accept-Language: en-GB or displayLanguage=en-GB

Your laptop is not a server: what not to copy

On your laptop	On a server
Ports on 127.0.0.1, no login	A gateway: TLS, one OAuth2 client per system, rate limits
Elasticsearch security off	Elasticsearch on a private network only; never a published port
Writable to load, then read-only	Read-only always; releases built and tested elsewhere
Version tag 11.0.0	Image pinned by digest, upgraded on purpose
One node, restart when you like	Two nodes or a cluster; warm the cache after restarts



Terima kasih · Questions

Docs: github.com/IHTSDO/snowstorm (docs/using-docker.md, docs/loading-snomed.md)

FHIR API: github.com/IHTSDO/snowstorm/blob/master/docs/using-the-fhir-api.md

ECL: docs.snomed.org/snomed-ct-specifications/snomed-ct-expression-constraint-language

Release notes: github.com/IHTSDO/snowstorm/releases (11.0.0, 2 Sept 2026)

Hands-on: Snowstorm-hands-on.postman_collection.json + environment file